

Metered Power Outlet

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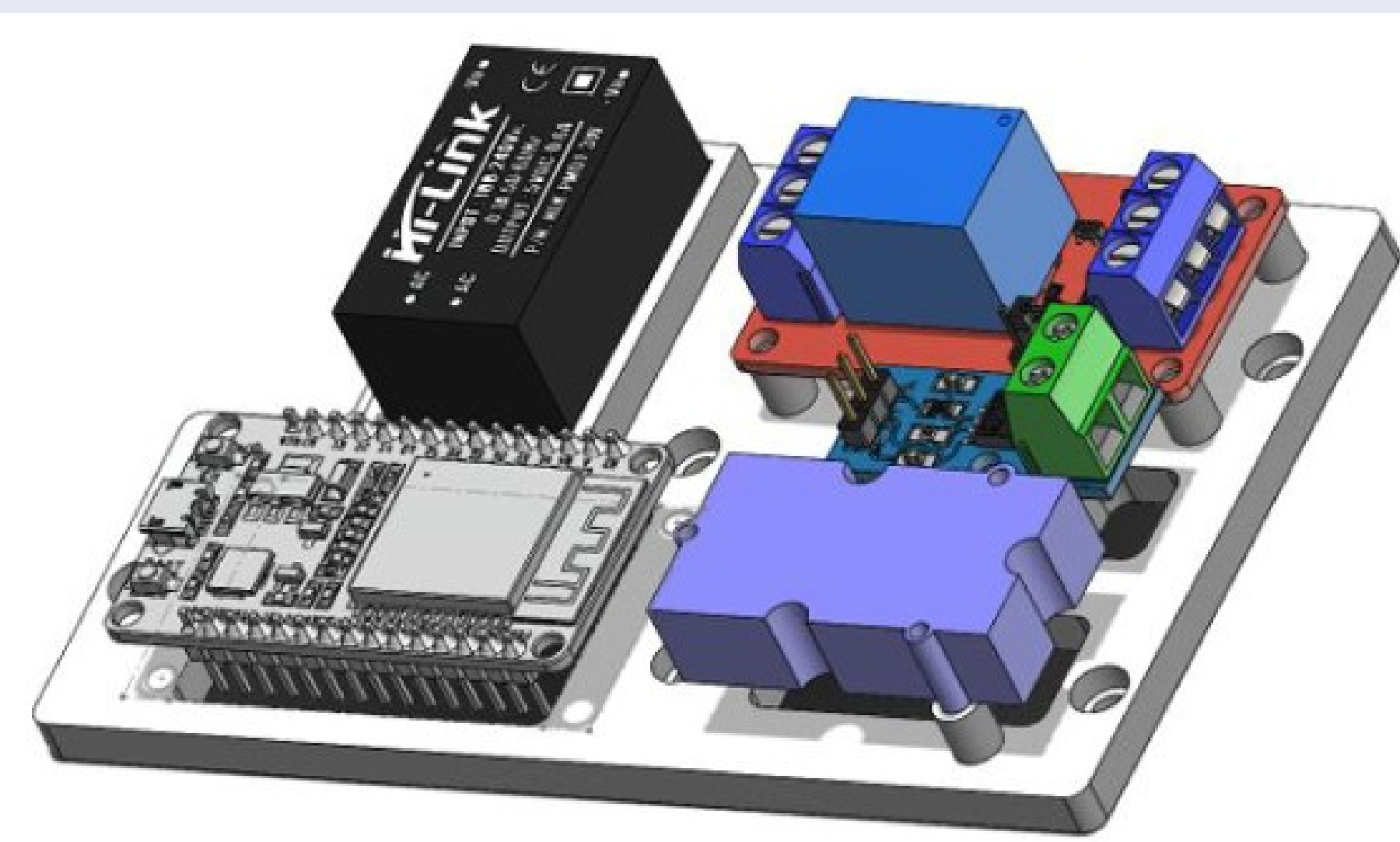
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Problem/Situation

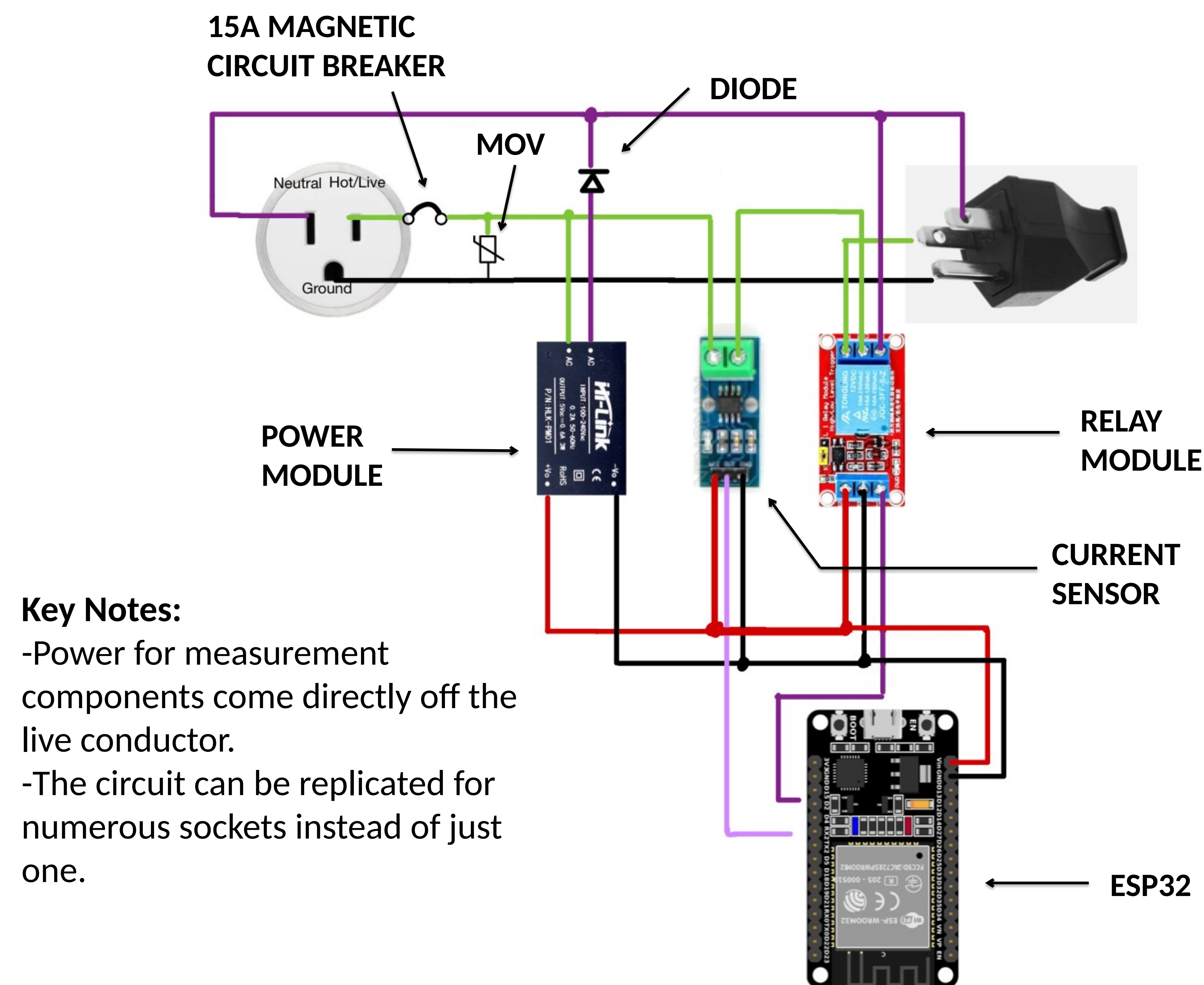
Consumers utilize electronics everyday, but we do not have a convenient way to monitor the electricity consumption of these electronics. Our project addresses this problem by giving users a tool to accurately measure voltage and current and report to the user how much electricity their electrical devices are using at any given time. Our product achieves this with a system of components.



Design Features

- Microcontroller: the “brains” of our product is the ESP32 microcontroller.
- Current sensor: instantaneous measurements of current.
- Relay module: handles termination of power to the applied load.
- User-friendly website: we provide the usage of power over time to the user via a user-friendly website to track power consumption of what ever device is plugged in.
- Make the circuit easy to replicate for the average end-user.

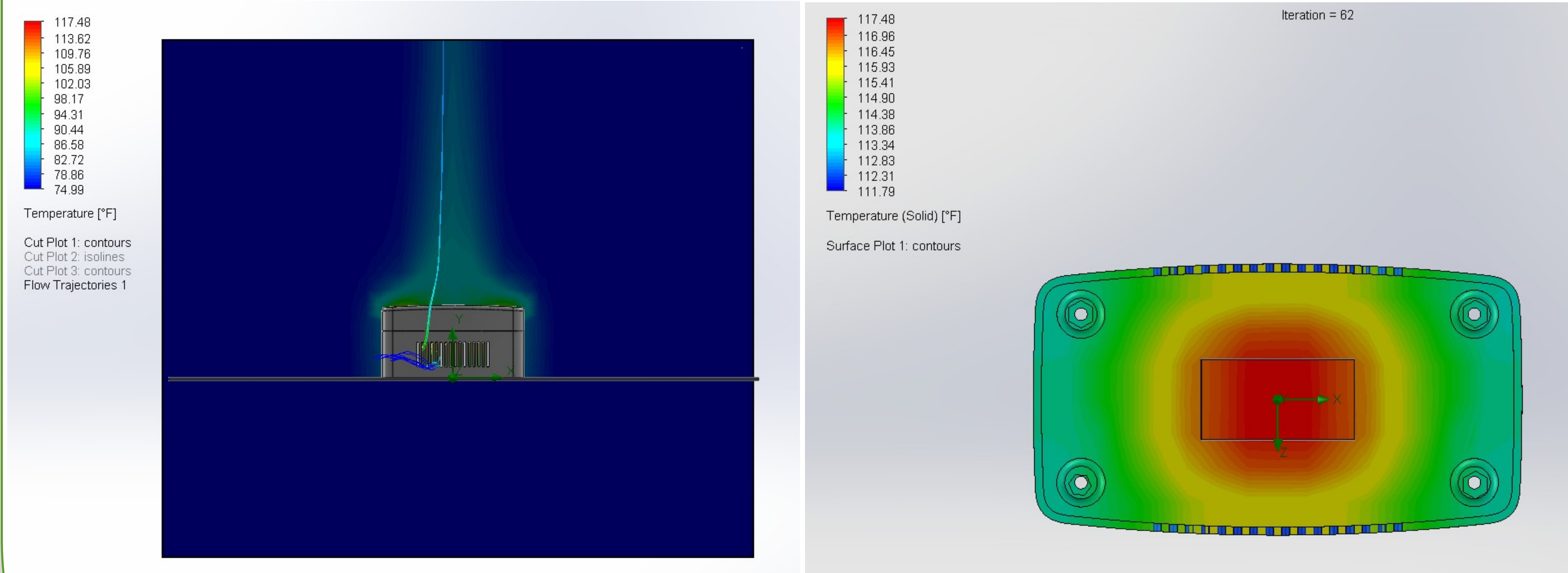
Circuit Diagram



Key Notes:

- Power for measurement components come directly off the live conductor.
- The circuit can be replicated for numerous sockets instead of just one.

Heat Analysis



Results



Challenges

- Implementing the safe measurements of AC power and all its differing attributes from well-versed DC power.
- Powering all our components for metering by tapping directly off the main circuit.
- Crafting our product to have a small footprint.

Next Generation Improvements

- Create a metered power strip with individual measurements and controls for each socket.
- Use more precise sensors.

Thanks, Credits, Contact

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